

Tugas : Laporan Praktikum Mandiri 5

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```
from google.colab import drive
drive.mount('/content/gdrive')

Mounted at /content/gdrive

path = "/content/gdrive/MyDrive/machine learning/praktikum/praktikum5"

import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv(path + '/data/Iris.csv')
df.head()
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

- Menghubungkan ke google
- Membuat path
- Mengimport modul untuk digunakan
- Menampilkan 5 data pertama

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 150 entries, 0 to 149  
Data columns (total 6 columns):  
#   Column          Non-Null Count  Dtype  
---  ---  
0   Id               150 non-null   int64  
1   SepalLengthCm    150 non-null   float64  
2   SepalWidthCm     150 non-null   float64  
3   PetalLengthCm    150 non-null   float64  
4   PetalWidthCm     150 non-null   float64  
5   Species          150 non-null   object  
dtypes: float64(4), int64(1), object(1)  
memory usage: 7.2+ KB
```

```
if 'Id' in df.columns:  
    df = df.drop('Id', axis=1)  
  
print("\nKolom dataset setelah dibersihkan:")  
print(df.columns)
```

```
Kolom dataset setelah dibersihkan:  
Index(['SepalLengthCm', 'SepalWidthCm', 'PetalLengthCm', 'PetalWidthCm',  
       'Species'],  
      dtype='object')
```

```
X = df.drop('Species', axis=1)  
y = df['Species']  
  
# Pembagian data 80% train dan 20% test  
X_train, X_test, y_train, y_test = train_test_split(  
    X, y, test_size=0.2, random_state=42, stratify=y  
)  
  
print("Jumlah data training:", X_train.shape[0])  
print("Jumlah data testing:", X_test.shape[0])
```

```
Jumlah data training: 120  
Jumlah data testing: 30
```

- Cek info dataset
- Data cleaning
- Memisahkan fitur (X) dan target (y), serta membagi data 80% train dan 20% test

```
model = DecisionTreeClassifier(criterion='gini', max_depth=4, random_state=42)
model.fit(X_train, y_train)
```

```
# Prediksi data testing
y_pred = model.predict(X_test)
```

```
print("Akurasi Model: {:.2f}%".format(accuracy_score(y_test, y_pred)*100))
```

```
print("\nConfusion Matrix:")
cm = confusion_matrix(y_test, y_pred)
print(cm)
```

```
print("\nClassification Report:")
print(classification_report(y_test, y_pred))
```

Akurasi Model: 93.33%

Confusion Matrix:

```
[[10  0  0]
 [ 0  9  1]
 [ 0  1  9]]
```

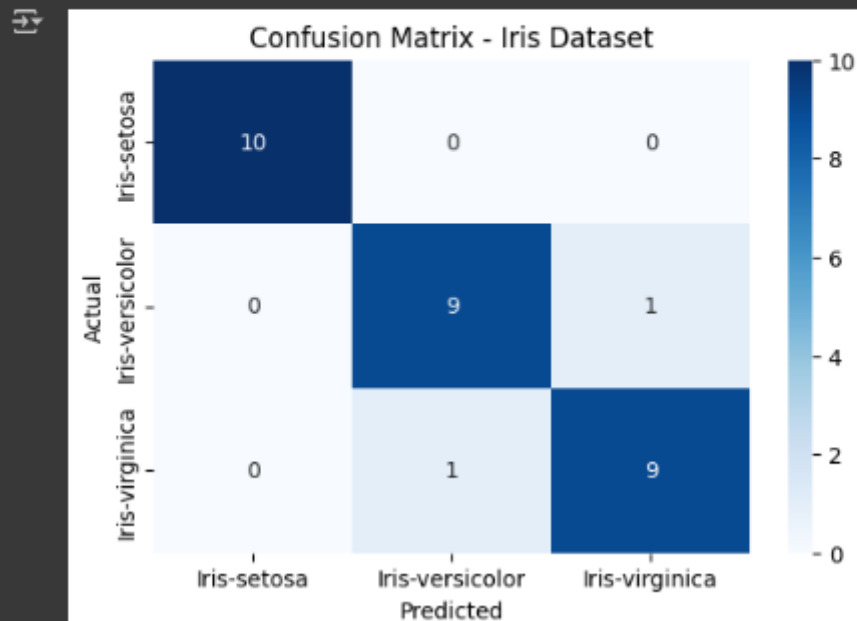
Classification Report:

	precision	recall	f1-score	support
Iris-setosa	1.00	1.00	1.00	10
Iris-versicolor	0.90	0.90	0.90	10
Iris-virginica	0.90	0.90	0.90	10
accuracy			0.93	30
macro avg	0.93	0.93	0.93	30
weighted avg	0.93	0.93	0.93	30

-Pembuatan model decision tree, serta prediksi data testing

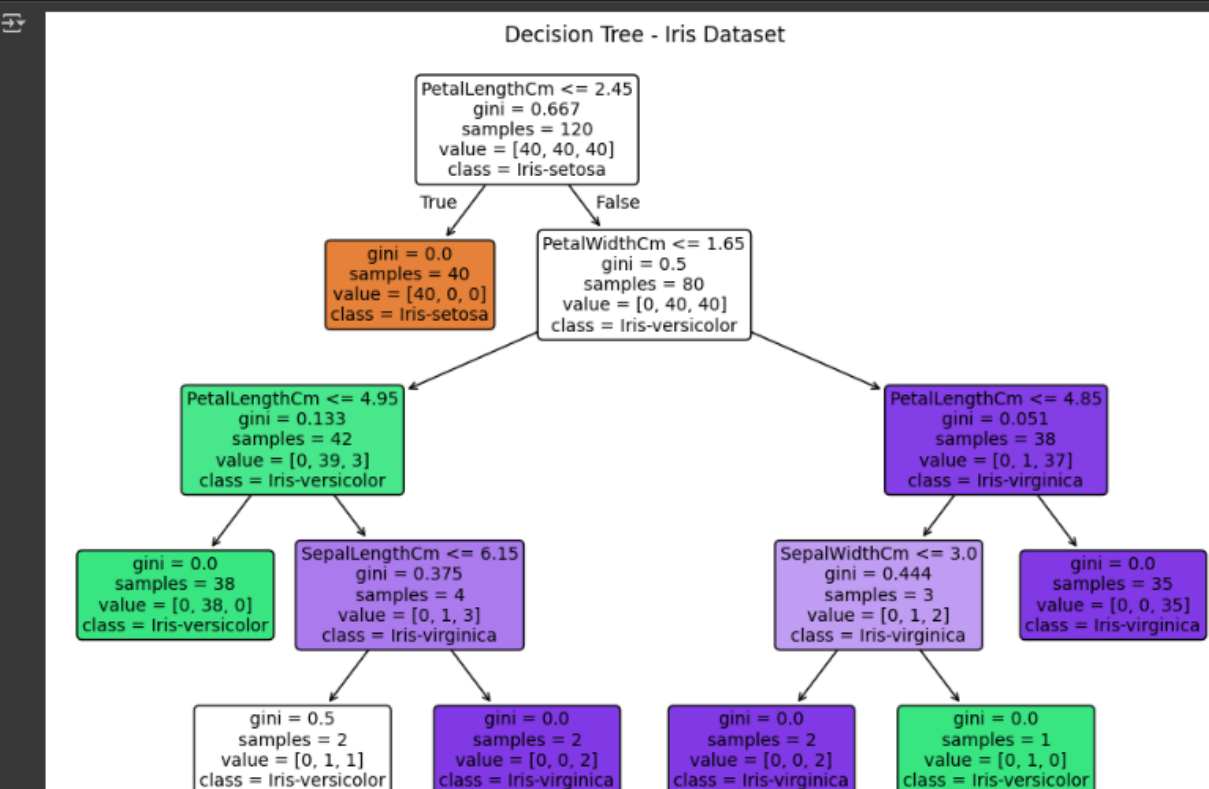
-Evaluasi model

```
plt.figure(figsize=(6,4))
sns.heatmap(cm, annot=True, cmap='Blues',
            xticklabels=model.classes_, yticklabels=model.classes_)
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix - Iris Dataset")
plt.show()
```



-Visualisasi confusion matrix

```
plt.figure(figsize=(12,8))
plot_tree(model, feature_names=X.columns, class_names=model.classes_,
          filled=True, rounded=True, fontsize=10)
plt.title("Decision Tree - Iris Dataset")
plt.show()
```



-Visualisasi decision tree

Link GitHub: <https://github.com/CeperMail/MachineLearning.git>